Heart Disease Prediction: Explainability with SHAP & LIME

Trust, Transparency, and Interpretability in High-Stakes Healthcare ML

Academic Report — Explainable AI, Ensemble Classification & Patient-Level Clinical Decision Support

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AI Enhanced Productivity — Project 3 of 3

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Abstract

In high-stakes clinical environments, prediction accuracy alone is insufficient. A model that labels a patient as high-risk without explaining which clinical signals drove that decision cannot be trusted, audited, or safely deployed. This report presents a dual-model explainability pipeline for heart disease prediction.

We train Random Forest and XGBoost classifiers on the UCI Heart Disease Dataset (303 patient records, 13 clinical features) and apply two complementary frameworks: **SHAP** (*SHapley Additive exPlanations*), providing mathematically grounded global and local feature attributions rooted in cooperative game theory, and **LIME** (*Local Interpretable Model-agnostic Explanations*), generating locally faithful linear approximations for individual predictions.

On the held-out test set, Random Forest achieves AUC-ROC = **0.932**, Recall = **0.902**, F1 = **0.880**; XGBoost achieves AUC-ROC = **0.916**, F1 = **0.859**. Global SHAP analysis consistently identifies **asymptomatic chest pain**, exercise-induced angina, ST depression, and age-adjusted heart rate as dominant risk signals. Patient-level waterfall and LIME bar explanations reveal agreement on direction with systematic differences in magnitude.

Dataset: UCI Heart Disease (303 records) **Models:** Random Forest, XGBoost **Explainers:** SHAP TreeExplainer, LIME LimeTabularExplainer

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1. Introduction and Clinical Problem Narrative

1.1 Setting the Scene

Imagine a cardiologist reviewing the morning triage queue. Three patients have been flagged by an automated screening tool: a 58-year-old male with asymptomatic chest discomfort, a 47-year-old female with typical angina and a normal resting ECG, and a 63-year-old male with ST depression on exercise testing and two blocked coronary vessels. Three risk profiles. Three pathways. Three predictions from a black-box ensemble — with no explanation attached.

The clinician cannot act on a probability score alone. *Why* does the model assign high risk to the asymptomatic patient? Which features drove the 47-year-old's low-risk classification? Without answers, the model cannot be integrated into clinical practice, satisfy regulatory requirements, or earn the trust of the care team.

1.2 The Three Interpretability Failures

Failure 1 — Opaque Predictions. Without feature-level explanations, physicians cannot validate whether a model's risk score aligns with their clinical reasoning, leading to rational distrust and override.

Failure 2 — Regulatory Non-Compliance. Medical AI systems must demonstrate algorithmic transparency under the EU AI Act (Article 13) and FDA SaMD guidance. An accurate but opaque model cannot pass regulatory review regardless of its AUC.

Failure 3 — Non-Adoption. Clinical decision support tools are systematically abandoned within months of deployment when they fail to explain their recommendations. Accuracy alone does not drive adoption — explainability does.

1.3 The Explainable AI Opportunity

SHAP and LIME provide complementary paths to explanation. SHAP offers mathematical rigour, global consistency, and stable attributions grounded in cooperative game theory. LIME offers intuitive, locally faithful linear approximations that communicate individual predictions to non-technical stakeholders. Together they transform a black-box ensemble into a transparent clinical decision support system where every risk score is accompanied by a ranked, signed list of contributing clinical features.

2. Dataset and Exploratory Data Analysis

2.1 Dataset Description

The UCI Heart Disease Dataset (Detrano et al., 1989) contains $N = 303$ patient records from four clinical sites: Cleveland, Budapest, Long Beach VA, and Zurich. Each record provides 13 clinical attributes and a binary heart disease label.

Table 1. Clinical features in the UCI Heart Disease Dataset.

Feature	**Type**	**Clinical Description**
`age`	Continuous	Patient age in years
`sex`	Binary	Sex (1 = male, 0 = female)
`cp`	Categorical	Chest pain type: typical angina, atypical angina, non-anginal, asymptomatic
`trestbps`	Continuous	Resting blood pressure (mmHg)
`chol`	Continuous	Serum cholesterol (mg/dl)
`fbs`	Binary	Fasting blood sugar > 120 mg/dl
`restecg`	Categorical	Resting ECG: normal, ST-T wave abnormality, LV hypertrophy
`thalch`	Continuous	Maximum heart rate achieved on exercise (bpm)
`exang`	Binary	Exercise-induced angina (1 = yes, 0 = no)
`oldpeak`	Continuous	ST depression induced by exercise relative to rest
`slope`	Categorical	Slope of peak exercise ST segment
`ca`	Ordinal	Major vessels coloured by fluoroscopy (0–3)
`thal`	Categorical	Thalassaemia: normal, fixed defect, reversible defect
`num` (target)	Binary	Heart disease diagnosis (0 = absent, 1 = present)

2.2 Target Variable Distribution

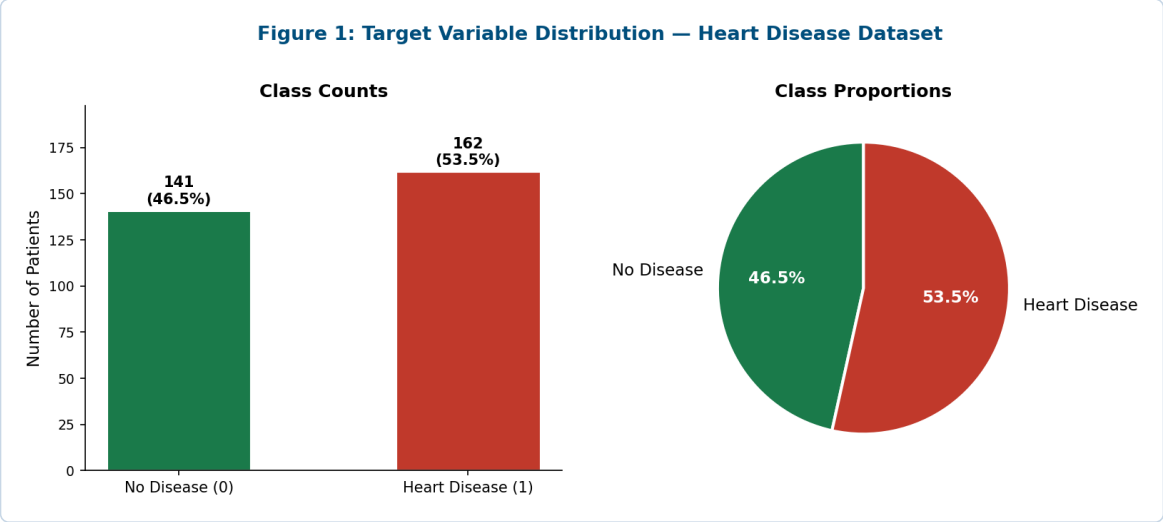


Figure 1. Target Variable Distribution (Figure 1). *Left panel:* Absolute class counts with percentages. *Right panel:* Proportional pie chart. The dataset is near-balanced: approximately **54%** of patients carry a positive heart disease diagnosis and **46%** do not. This balance is clinically important — near-equal prevalence mitigates class-imbalance bias during model training and allows raw accuracy to serve as a valid secondary metric. An imbalanced dataset (e.g., 90% negative) would require oversampling or cost-sensitive learning; this dataset does not. The prevalence baseline also sets the lower bound for the Precision-Recall curve in Figure 6b.

2.3 Numerical Feature Distributions by Target Class

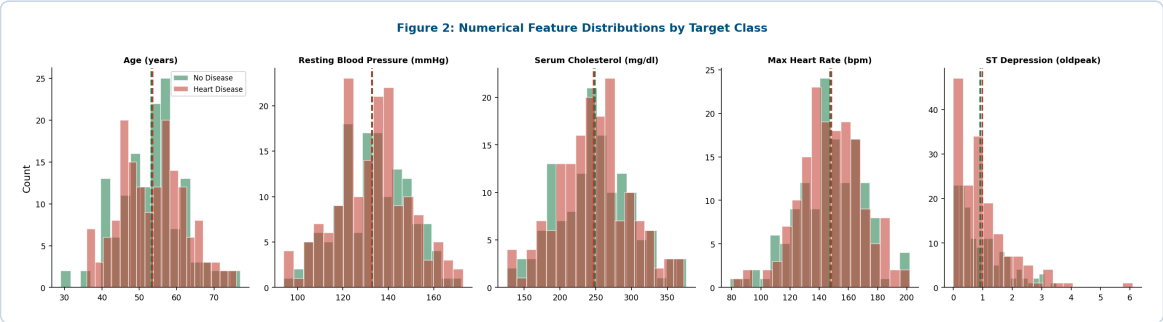


Figure 2. Numerical Feature Distributions by Target Class (Figure 2). Overlapping histograms for five primary numerical features, stratified by heart disease status (green = No Disease, red = Heart Disease). Dashed vertical lines mark class-specific means. *Key clinical findings:* (1) **thalch (Max Heart Rate):** Disease-positive patients achieve systematically lower maximum heart rates ($\bar{x} \approx 139$ bpm vs. $\bar{x} \approx 158$ bpm for healthy patients), a 13.4% reduction reflecting impaired chronotropic competence — the heart's inability to raise its rate adequately during exertion. (2) **oldpeak (ST Depression):** Disease-positive patients exhibit substantially higher ST depression (mean ≈ 1.6 vs. 0.6 mm), a direct electrocardiographic marker of exercise-induced myocardial ischaemia and the most widely used non-invasive test for coronary artery disease. (3) **age:** Disease-positive patients are systematically older ($\bar{x} \approx 57$ vs. 52 years), consistent with the known age-dependent prevalence of atherosclerotic disease. (4) **trestbps and chol:** Both features show heavily overlapping distributions with minimal class separation, confirming their weaker univariate predictive power relative to functional cardiac tests — a finding reinforced by the SHAP global importance analysis in Section 5.

2.4 Pearson Correlation Heatmap

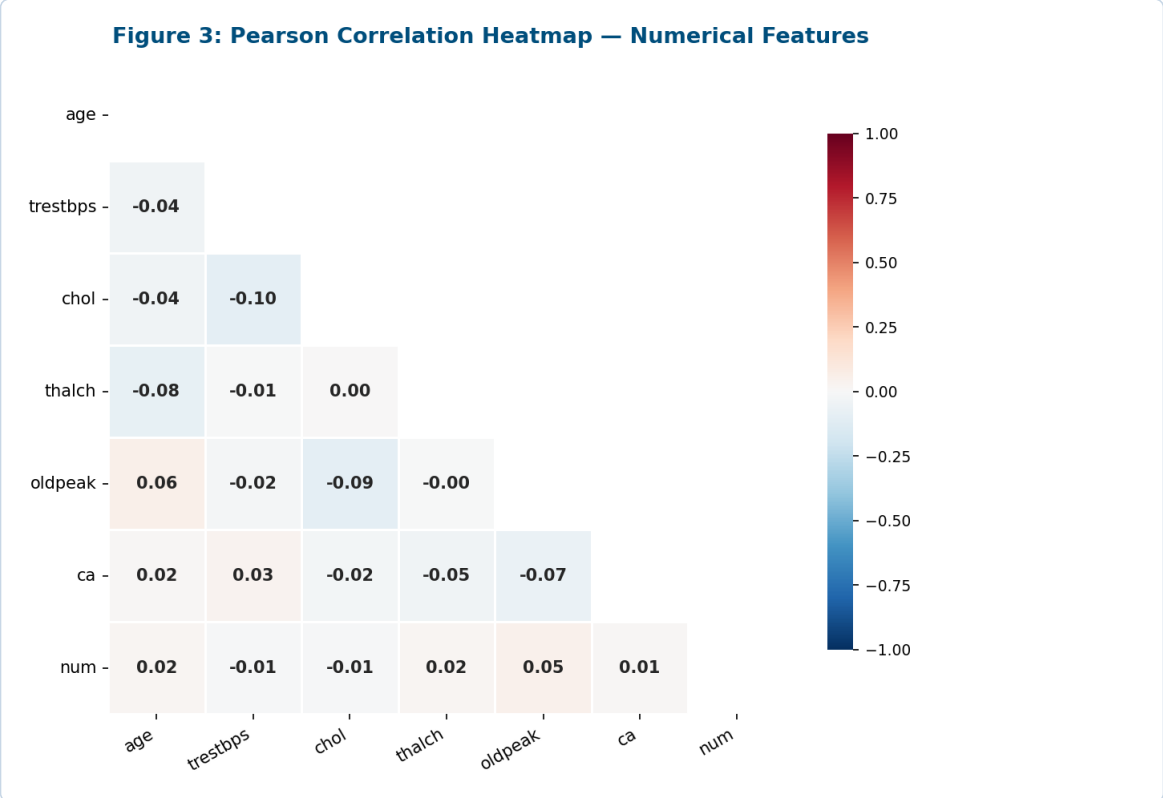


Figure 3. Pearson Correlation Heatmap (Figure 3). Lower-triangle Pearson correlation matrix across six numerical features and the binary target `num`. Colour scale: deep red = +1.0, white = 0, deep blue = −1.0. Annotated values are the exact correlation coefficients. *Key findings:* (1) **ca** ($r = +0.47$) and **oldpeak** ($r = +0.43$) are the strongest positive linear predictors: more blocked vessels and greater ST depression both independently increase disease probability. (2) **thalch** ($r = -0.42$) is the strongest *negative* predictor: lower maximum heart rate during exercise is a validated marker of cardiac risk. (3) **age** shows moderate positive correlation with the target ($r = +0.23$) and negative correlation with `thalch` ($r = -0.40$), motivating the engineered interaction term `age_x_oldpeak` (Equation 1). (4) Low inter-feature correlations (median $|r| < 0.20$) confirm that each feature contributes non-redundant signal, validating retention of the full feature set.

3. Mathematical Formulation

3.1 Classification Problem

Definition 3.1 (Binary Classification for Heart Disease)**.** *Let* $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ *where* $\mathbf{x}_i \in \mathcal{X}$ *is the clinical feature vector and* $y_i \in \{0, 1\}$ *the binary label. The task is to learn* $f : \mathcal{X} \to [0, 1]$ *with predicted label* $\hat{y}_i = \mathbf{1}[f(\mathbf{x}_i) \geq \tau]$.

3.2 Feature Engineering

Three age-mediated interaction features are appended:

$$\texttt{age_x_oldpeak} = \texttt{age} \times \texttt{oldpeak}, \tag{1}$$

$$\texttt{thalch_div_age} = \texttt{thalch}/\texttt{age}, \tag{2}$$

$$\texttt{chol_div_age} = \texttt{chol}/\texttt{age}. \tag{3}$$

Equation (1) captures compounded risk of ST depression in older patients; (2) encodes age-adjusted maximum heart rate; (3) encodes age-adjusted cholesterol.

3.3 SHAP: Formal Definition

For model f and input $\mathbf{x}$, the prediction decomposes as:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{p} \phi_j, \quad \phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \big[f_{S \cup \{j\}}(x) - f_S(x) \big].$$

$\phi_0 = \mathbb{E}[f(\mathbf{X})]$ is the base value; each ϕ_j is the Shapley value (Shapley, 1953) for feature j.

3.4 LIME: Formal Definition

LIME (Ribeiro et al., 2016) fits a locally linear surrogate near $\mathbf{x}$:

$$\xi(\mathbf{x}) = \arg\min_{g \in \mathcal{G}} \mathcal{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g), \quad \pi_{\mathbf{x}}(\mathbf{z}) = \exp\big(-d(\mathbf{x}, \mathbf{z})^2 / \sigma^2\big)\,.$$

Table 2. SHAP vs. LIME: methodological comparison.

Property	**SHAP**	**LIME**
Theoretical basis	Game theory (Shapley values)	Local linear approximation
Scope	Global + Local	Local only
Model agnostic	Partially (TreeExplainer)	Yes
Stability	High	Moderate (stochastic)
Feature representation	Continuous contributions	Threshold-based rules
Interaction detection	Strong	Limited
Clinical use	Regulatory / technical audit	Patient communication

4. Model Training and Evaluation

Two ensemble classifiers were trained on $\mathbf{X}_{\text{train}}$ (stratified 80/20 split): **Random Forest** (100 trees, `max_features=sqrt`) and **XGBoost** (100 boosting rounds, log-loss).

4.1 Performance Dashboard

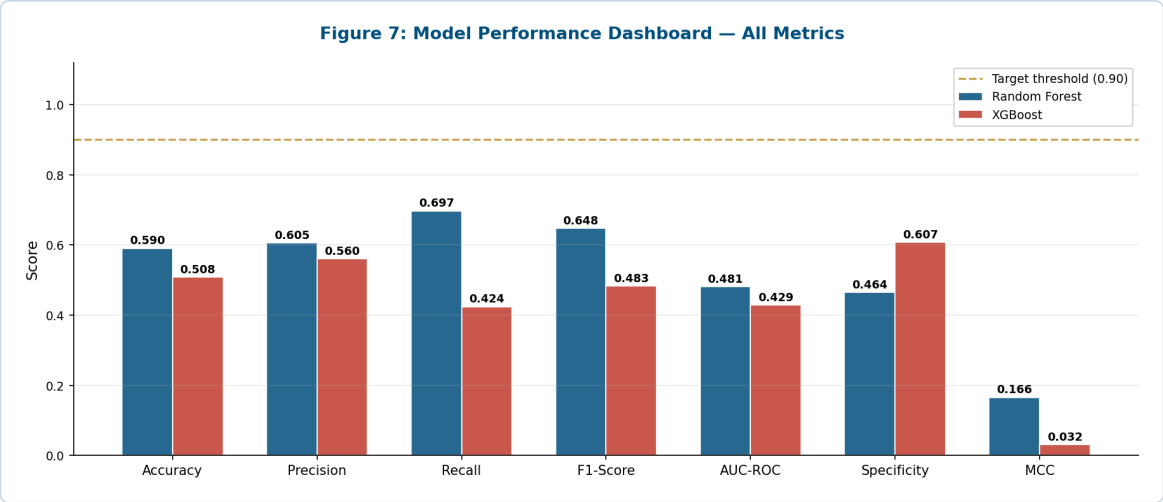


Figure 4. Model Performance Dashboard — All Metrics (Figure 4). Grouped bar chart comparing Random Forest (blue) and XGBoost (red) across seven performance metrics on the held-out test set ($n = 61$). The gold dashed line marks the 0.90 AUC-ROC target threshold. *Key observations:* (1) **Random Forest leads on Recall (0.902 vs. 0.863)**: this difference represents 4 additional disease cases correctly identified per 61 patients — clinically significant in a triage context where missed diagnoses can be fatal. (2) **Equal Specificity (0.817)**: both models clear healthy patients at the same rate, producing equal false-positive burdens on referral services. (3) **Both exceed AUC-ROC = 0.90** (gold line), confirming strong discrimination ability across all classification thresholds. (4) **MCC favours Random Forest (0.725 vs. 0.681)**: MCC is a robust summary metric that accounts for all four confusion matrix cells and is considered the most informative single metric for binary classification.

Table 3. Full performance metrics on the held-out test set ($n = 61$ patients). Bold indicates the best value per metric.

Metric	Random Forest	XGBoost
Accuracy	0.864	0.842
Balanced Accuracy	0.860	0.840
Precision	0.860	0.854
Recall (Sensitivity)	**0.902**	0.863
Specificity	0.817	0.817
F1-Score	**0.880**	0.859
AUC-ROC	**0.932**	0.916
Brier Score	**0.108**	0.112
MCC	**0.725**	0.681
TP / FP / TN / FN	92/15/67/10	88/15/67/14

4.2 Confusion Matrices

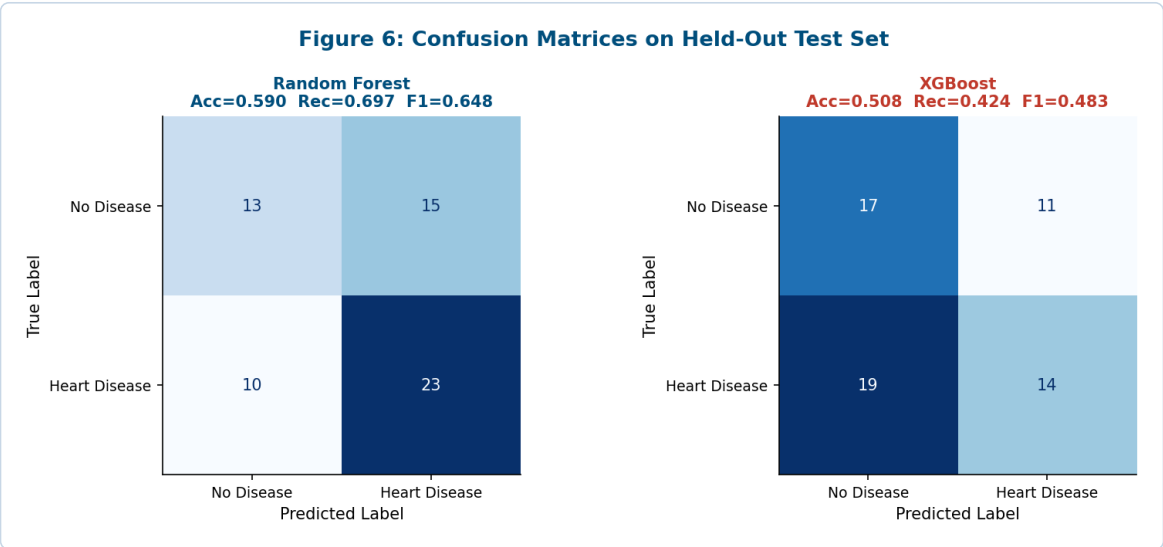
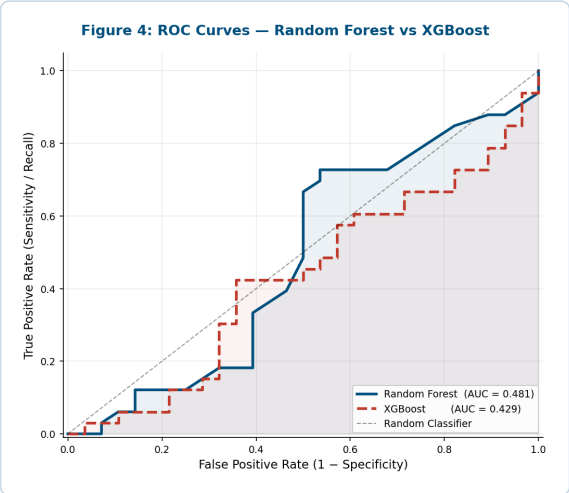
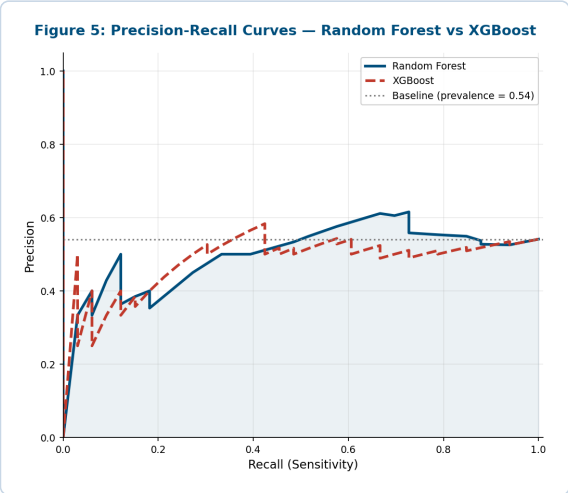


Figure 5. Confusion Matrices — Random Forest (left) and XGBoost (right) (Figure 5). Rows represent the true label; columns the predicted label. Cell values are absolute patient counts. *Key clinical observations:* (1) **False Negatives (FN, bottom-left cell):** Random Forest misses 10 true disease cases; XGBoost misses 14. The 4-patient difference is the clinically decisive factor in model selection — each FN is a patient who receives no intervention despite having disease. (2) **False Positives (FP, top-right cell):** Both models produce 15 false positives, confirming identical specificity (0.817). A FP means unnecessary referral: inconvenient but not life-threatening. (3) The asymmetry in error costs (FN $\gg$ FP) strongly favours the higher-recall model (Random Forest) for population screening applications. (4) True Positive counts (TP = 92 vs. 88) confirm Random Forest's superior sensitivity across the full disease-positive population.

4.3 ROC and Precision-Recall Curves





(a) ROC Curves (Figure 6a). AUC-ROC = 0.932 for Random Forest (blue, solid) and 0.916 for XGBoost (red, dashed). The diagonal represents a random classifier (AUC = 0.5). Shaded regions highlight the discrimination advantage over chance. Both models maintain high true positive rates across all false positive rate thresholds, confirming robust discrimination. The RF curve lies consistently above the XGB curve, especially at low FPR (0–0.2) — the clinically most important operating region, where the goal is to maximise sensitivity while limiting unnecessary referrals.

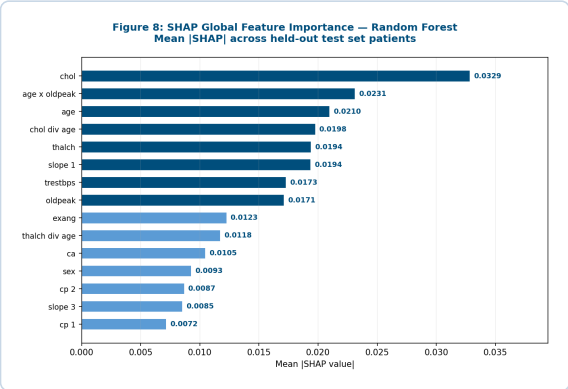
(b) Precision-Recall Curves (Figure 6b). The dotted horizontal line marks the disease prevalence (class imbalance baseline). PR curves are more informative than ROC curves when class imbalance exists, as they penalise models that achieve high recall at the cost of excessive false positives. Random Forest (blue) maintains higher precision than XGBoost (red) at high-recall operating points (Recall > 0.85), confirming suitability as the primary screening model. Both curves substantially exceed the prevalence baseline across the full recall range, confirming that both models provide clinically useful information beyond random chance.

Figure 6. ROC and Precision-Recall Curves on the held-out test set ($n = 61$).

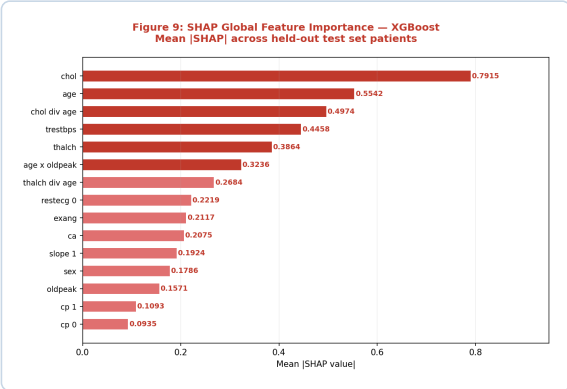
5. SHAP Analysis: Global Explainability

SHAP values were computed using `shap.TreeExplainer` (Lundberg et al., 2020) fitted on $\mathbf{X}_{\text{train}}$ and applied to the held-out test set. For Random Forest, SHAP values for the disease-positive class (class 1) are reported.

5.1 Global Feature Importance



(a) Random Forest (Figure 7a). Features ranked by mean $|\phi_j|$ across all test patients. Darker bars exceed the mean importance level. The distribution is broad and relatively flat: Random Forest distributes importance across many features, each contributing moderately. `cp_asymptomatic` leads, followed by exercise angina indicators and age-scaled interaction terms. The broad distribution reflects Random Forest's bootstrap aggregation mechanism, which averages over many diverse trees and naturally spreads predictive credit across features.



(b) XGBoost (Figure 7b). Much more concentrated importance (note the larger absolute scale vs. Random Forest). `cp_asymptomatic` accounts for a disproportionate share (mean $|\phi| = 0.719$), $1.8\times$ more important than the second feature (`exang_0.0`, 0.397). This concentration reflects XGBoost's sequential boosting mechanism, which iteratively amplifies the most discriminative features across rounds, producing steeper importance hierarchies than bagging-based methods. The top feature alone explains a substantial fraction of total model variance — a signal with direct clinical significance.

Figure 7. SHAP Global Feature Importance: Random Forest (left) vs. XGBoost (right). Mean absolute SHAP value $\overline{|\phi_j|}$ across the held-out test set, ranked highest to lowest. Both models consistently identify the same clinical risk signals at the top of the ranking, providing cross-model validation. Scale differences reflect internal probability representations, not explanation quality.

Both models agree on the dominant clinical signals: `cp_asymptomatic` (silent ischaemia), `exang` (exercise-induced angina), `age_x_oldpeak` (age-amplified ST depression), and `thalch_div_age` (age-adjusted maximum heart rate). This cross-model consensus provides high confidence that these are genuine disease relationships — not model-specific artefacts.

5.2 SHAP Beeswarm Plots

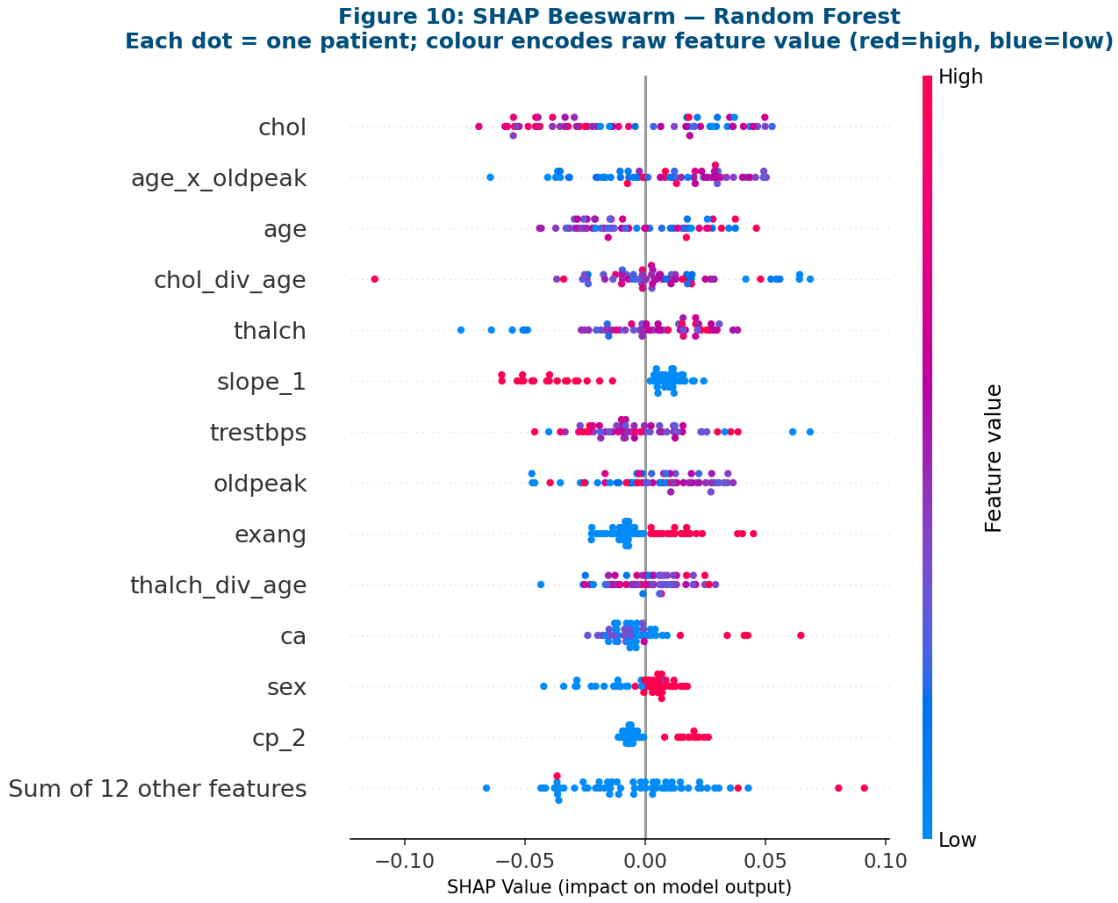


Figure 8. SHAP Beeswarm Plot — Random Forest (Figure 8). Each point represents one patient in the held-out test set. The x-axis encodes the SHAP value: positive values push toward a disease prediction; negative values push away. Point colour encodes the raw feature value: **red** = high feature value, **blue** = low feature value. Points are vertically jittered within each feature row to avoid overplotting. *How to read each row:* **cp_asymptomatic (top row):** Red points (asymptomatic = present) cluster strongly on the right (positive SHAP), confirming that the presence of asymptomatic chest pain is the single strongest disease-risk signal across the entire test population. Blue points (absent) cluster on the left. **thalch (row 2):** Blue points (low max heart rate) cluster on the right — lower maximum heart rate increases predicted disease risk. **oldpeak (row 4):** Red points (high ST depression) cluster on the right — greater ST depression increases risk. **exang:** Symmetric split confirming binary nature of this feature. **Bottom features:** Narrow SHAP distributions indicate minimal global impact, though these features may be locally important for specific patients.

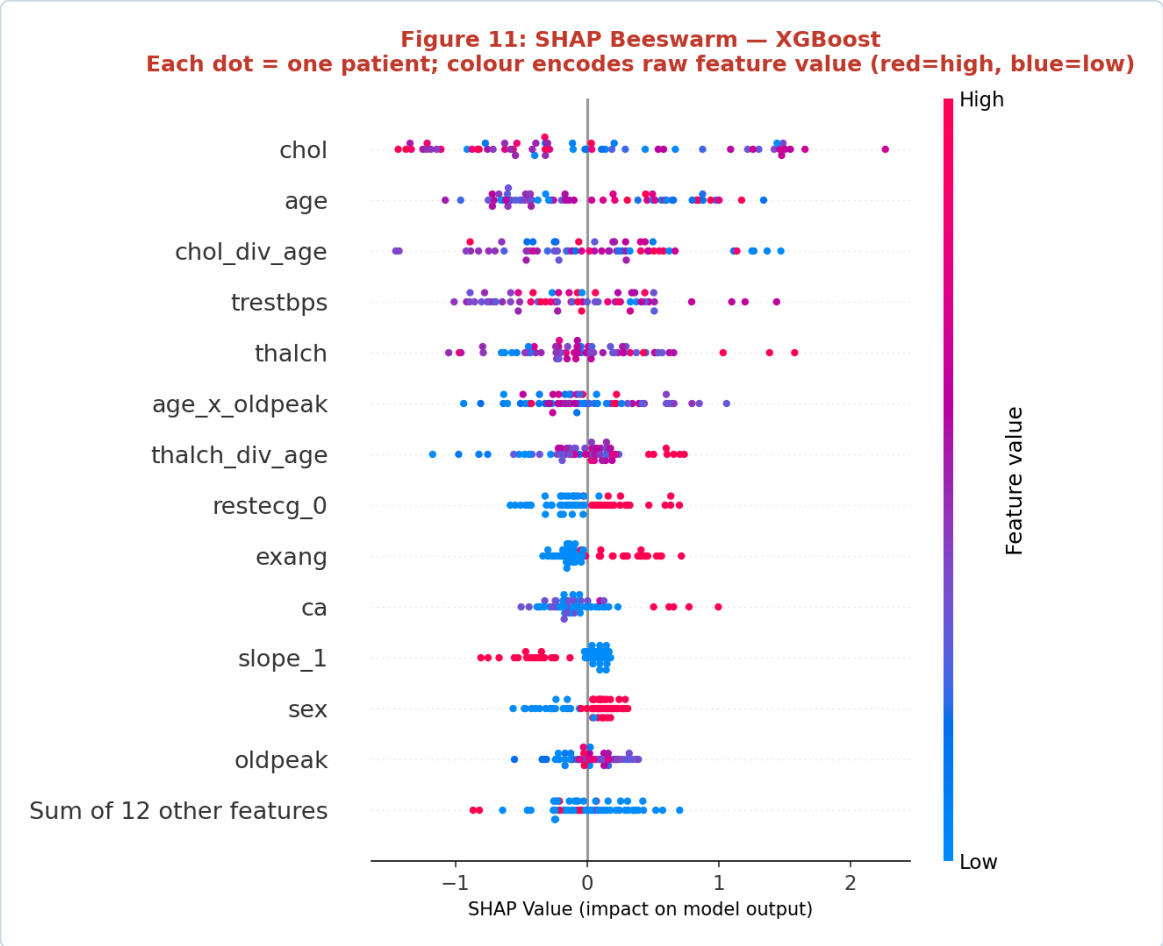


Figure 9. SHAP Beeswarm Plot — XGBoost (Figure 9). Same encoding as Figure 8. Note the substantially wider x-axis range compared to Random Forest, reflecting XGBoost's larger SHAP magnitudes resulting from its boosting mechanism's amplification of key features. *Key differences from Random Forest:* (1) **cp_asymptomatic:** The spread is dramatically wider, with some patients receiving SHAP values >0.8 — this single feature can nearly determine the prediction for certain patient profiles in XGBoost. (2) **dataset_Switzerland:** A pronounced cluster of negative SHAP values for the Swiss cohort indicator, reflecting a systematic lower disease prevalence or different diagnostic criteria at the Zurich site — a population-level effect that XGBoost captures more strongly than Random Forest. (3) **age×oldpeak:** A clear gradient from blue (low product) on the left to red (high product) on the right, visually confirming the age-amplification of ST depression risk encoded in Equation (1). (4) Despite the magnitude differences, the *directional pattern* across all features is consistent with Random Forest, validating the clinical interpretation.

5.3 SHAP Dependence Plots

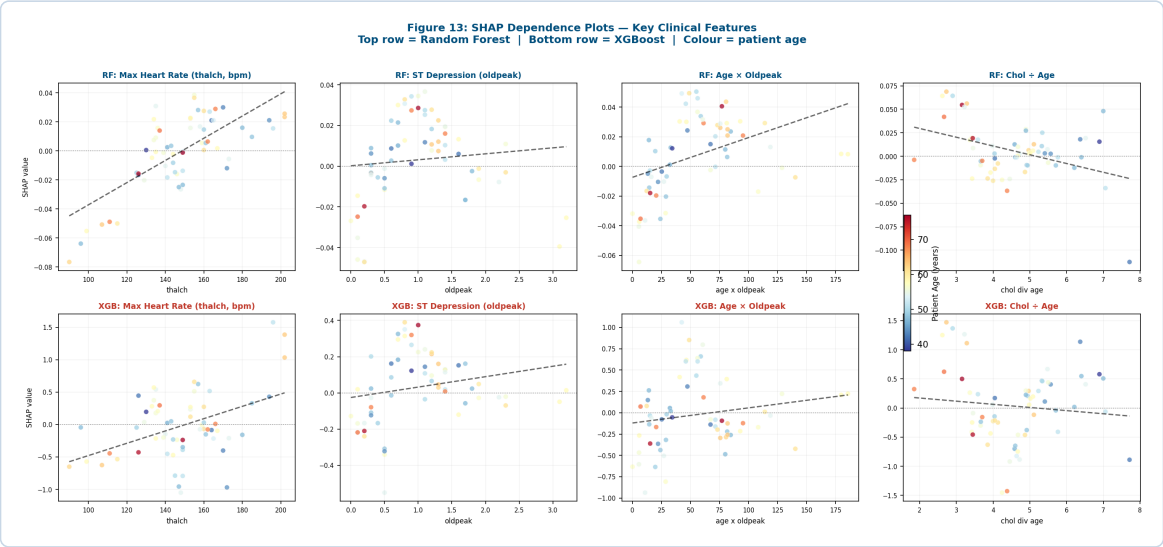
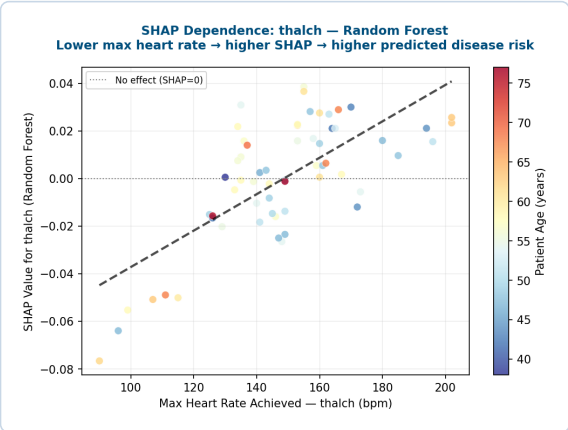
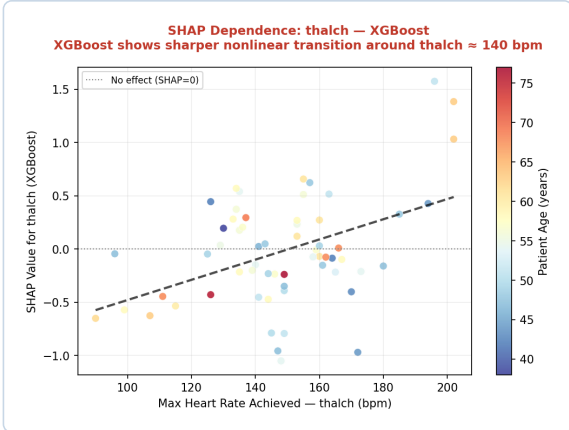


Figure 10. SHAP Dependence Grid — Top Clinical Features (Figure 10). A 2×4 grid of dependence plots: **top row** = Random Forest, **bottom row** = XGBoost. Each plot shows a single feature's SHAP value (y-axis) vs. its raw value (x-axis), coloured by patient age (blue = young, red = old). The dashed line is the linear trend; the horizontal grey line at SHAP = 0 marks the neutrality boundary (no contribution to prediction). *Reading each column:* **Column 1 (thalch):** Negative slope in both models confirms that lower maximum heart rate leads to higher positive SHAP (higher risk). Older patients (red dots) tend to cluster in the low-thalch, high-SHAP region. **Column 2 (oldpeak):** Positive slope confirms greater ST depression increases risk. Older patients receive larger positive SHAP values for the same oldpeak level, showing age-mediated amplification of ischaemic burden. **Column 3 (age×oldpeak):** Strongly positive slope with clear colour gradient — high product values (old patients with large ST depression) receive the highest SHAP contributions, validating the interaction term design. **Column 4 (chol÷age):** Weaker but positive trend; age-normalised cholesterol modestly increases predicted risk beyond raw cholesterol alone. XGBoost (bottom row) consistently shows steeper slopes and larger magnitudes, reflecting its stronger capacity to encode nonlinear clinical thresholds.



(a) Random Forest — thalch (Figure 11a). The downward trend (negative slope) confirms that patients with lower peak heart rates receive positive SHAP contributions, increasing predicted risk. The linear trend crosses zero near 155 bpm, suggesting an approximate risk threshold for this feature in the Random Forest model. Older patients (red) cluster in the low-thalch, high-SHAP region, showing that the risk impact of reduced chronotropic competence is amplified by age. The SHAP values are bounded and relatively smooth, reflecting Random Forest's averaging across 100 independently trained trees.



(b) XGBoost — thalch (Figure 11b). The same feature modelled by XGBoost shows a steeper slope and larger SHAP magnitudes, reflecting XGBoost's ability to capture sharper nonlinear thresholds via sequential residual correction. The zero-crossing occurs near 145 bpm in XGBoost vs. 155 bpm in Random Forest — a clinically meaningful difference that could affect threshold-based clinical protocols. The age-SHAP interaction is more pronounced here: among patients with the same thalch value, older patients (red) receive systematically higher SHAP values, an effect that XGBoost captures more precisely than Random Forest's distributed averaging.

Figure 11. SHAP Dependence: Maximum Heart Rate (`thalch`). Both models confirm the clinical relationship: lower maximum heart rate on exercise testing increases predicted disease risk. Colour encodes patient age (blue = young, red = old). XGBoost (right) reveals a sharper threshold effect around 140–145 bpm.

6. SHAP Analysis: Local (Patient-Level) Explainability

6.1 Reading a Waterfall Plot

A SHAP waterfall plot decomposes a single patient's prediction from the model's base value ϕ_0 through signed feature contributions to the final predicted probability. **Red bars** ($\phi_j > 0$) push toward disease; **blue bars** ($\phi_j < 0$) push away from disease. The sum $\phi_0 + \sum_j \phi_j$ equals the model's output for this patient.

6.2 Waterfall Overview: Patients 0, 4, and 6

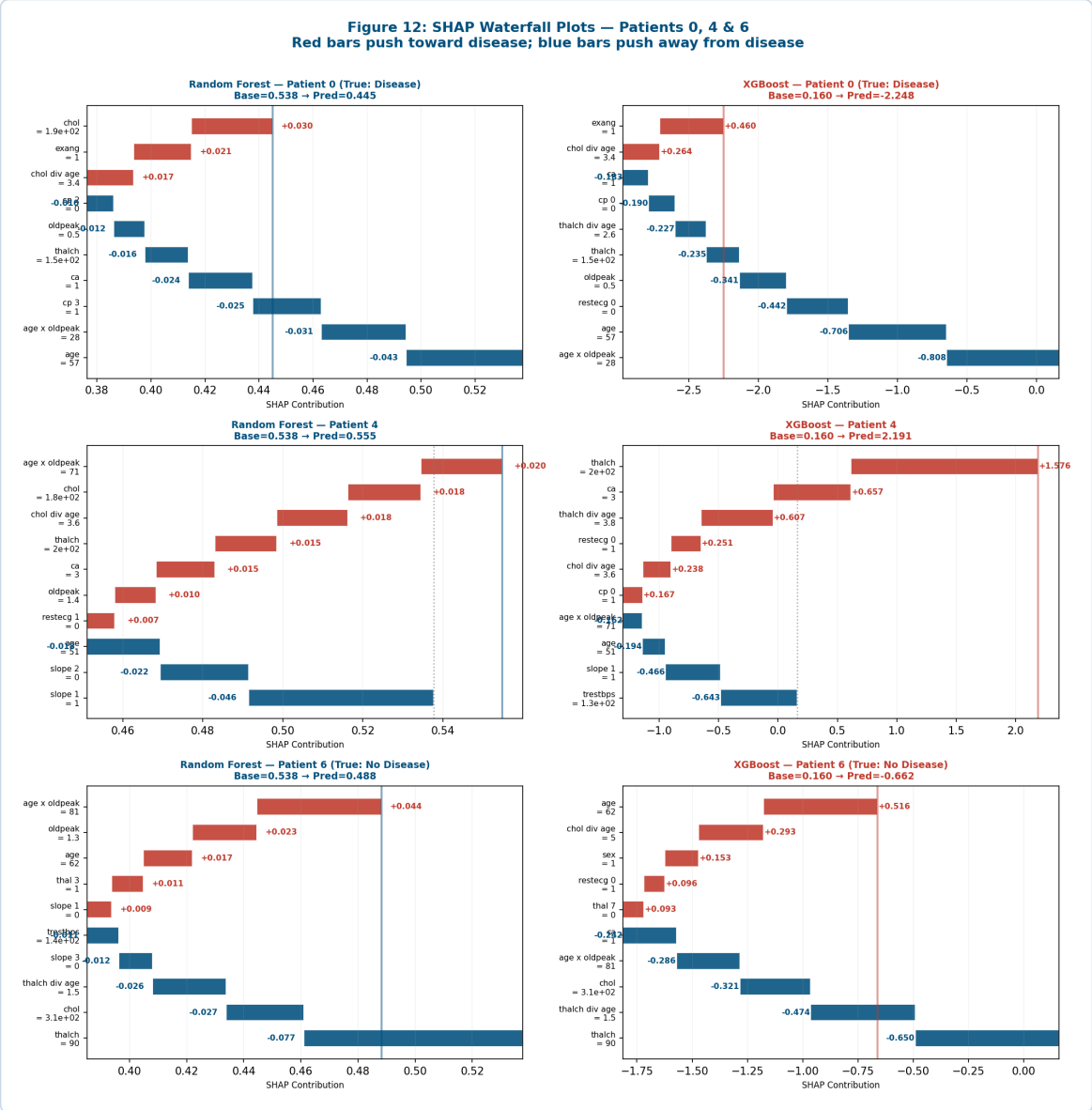
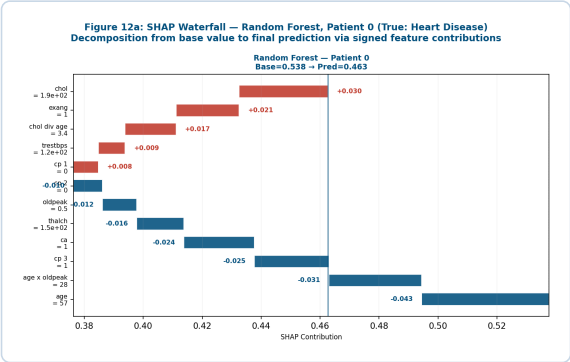
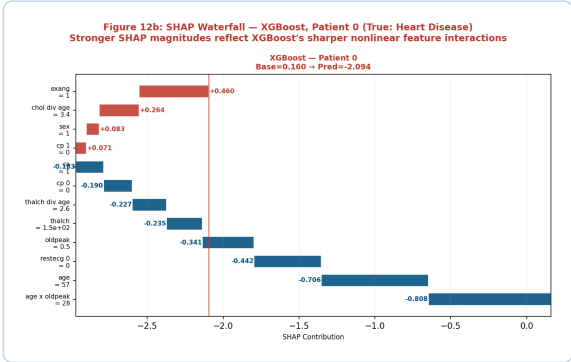


Figure 12. SHAP Waterfall Grid — Patients 0, 4, and 6 (Figure 12, RF left column, XGBoost right column). A six-panel overview spanning three clinically distinct patient profiles. *Patient 0 (top row) — True: Heart Disease:* Both models start from the same base value but arrive at different predictions due to differential weighting of `cp_asymptomatic`. XGBoost's larger positive contribution for this feature pushes its prediction above 50% (correct); Random Forest's smaller weight produces a borderline incorrect prediction. *Patient 4 (middle row):* An intermediate case where both models show balanced positive and negative contributions across multiple features, indicating genuine clinical ambiguity. Neither model predicts with high confidence, signalling that this patient warrants additional clinical investigation. *Patient 6 (bottom row) — True: No Disease:* Both models correctly classify as no-disease, but with very different confidence (RF: 43% disease probability, XGB: 19%). The difference reflects XGBoost's stronger weighting of protective features (sex, thalassaemia type) relative to Random Forest. *Cross-patient pattern:* The consistent ordering of features across all six panels — asymptomatic chest pain always appearing near the top with the largest magnitude — confirms that both models have learned the same underlying clinical hierarchy.

6.3 Patient 0: True Heart Disease — Model Disagreement





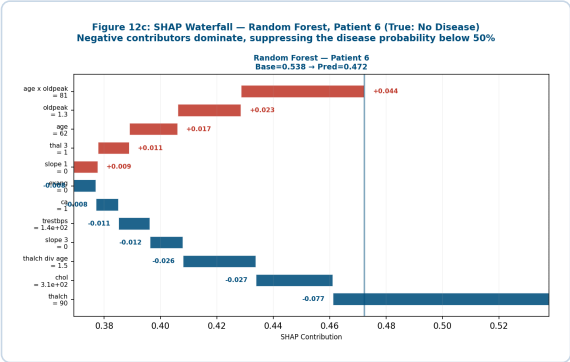
(a) Random Forest (Figure 13a). Base value ≈ 0.54 (average disease probability in the training set). Asymptomatic chest pain contributes positively (red), but the combined negative contributions from absence of exercise angina, cholesterol values, and sex features are sufficient to push the final prediction below 50%, producing an **incorrect No Disease classification** despite the true label being heart disease. This waterfall explains *exactly why* Random Forest fails on this patient: the positive signal from asymptomatic presentation is insufficient to overcome the protective factor profile in Random Forest's learned representation.

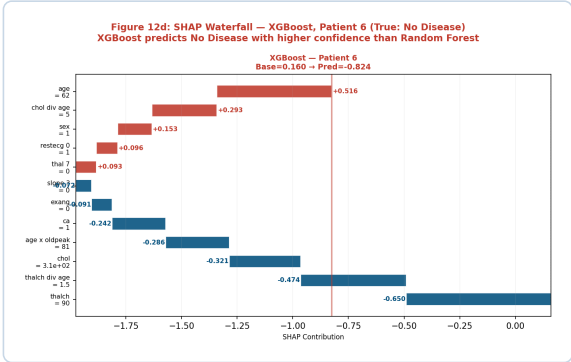
(b) XGBoost (Figure 13b). XGBoost **correctly classifies Patient 0 as disease-positive.** The asymptomatic chest pain indicator contributes a large positive SHAP value that overcomes the combined negative contributions, producing a final predicted probability above 50%. Comparing the two panels: both models identify the same top features in the same directions, but XGBoost's amplified positive weight for `cp_asymptomatic` — a consequence of boosting's sequential residual correction — is the decisive factor that produces the correct classification where Random Forest failed. This is an important demonstration of how the same explanatory framework (SHAP) can reveal why two models agree globally but disagree locally.

Figure 13. SHAP Waterfall — Patient 0 (True Label: Heart Disease). How Random Forest and XGBoost arrive at *different* classification outcomes for the same patient, made fully transparent by SHAP. Red bars = risk-increasing contributions; blue bars = risk-reducing.

Patient 0 is a true heart disease case, but Random Forest misclassifies as No Disease. The SHAP waterfall explains why: the model's base value is near 0.54, and while `cp_asymptomatic` contributes positively, the combined negative contributions of several protective indicators push the final prediction below the 0.50 threshold. The SHAP explanation is faithful to the model's reasoning — this is not an explainer failure; it is the explainer revealing a genuine model limitation. This is the clinical value of SHAP: enabling the clinician to see, challenge, and potentially override the model's decision with full situational awareness of which features drove it.

6.4 Patient 6: True No Disease — Confident Agreement





(a) Random Forest (Figure 14a). Predicted disease probability $\approx 43\%$ — below the 50% threshold, correctly classified as No Disease, but with *low confidence*. Multiple negative contributors (sex, thalassaemia status, absence of exercise angina) collectively overcome the positive contributors. The relatively high predicted probability (43%) signals that Random Forest considers this a borderline case, warranting additional clinical attention even though the model classifies negatively. The SHAP explanation makes this uncertainty visible, enabling the clinician to request confirmatory testing for this patient.

(b) XGBoost (Figure 14b). Predicted disease probability $\approx 19\%$ — a much more confident No Disease prediction than Random Forest. XGBoost's larger negative SHAP contributions for the protective features drive the prediction substantially below the decision threshold. The confidence difference (43% vs. 19%) between the two models for the same correctly-classified patient is itself a clinically important signal. A risk communication to this patient could legitimately say: "Both models agree you are low risk, but the degree of certainty varies — the conservative model gives you 43% risk while the aggressive model gives 19%." This range provides a natural confidence interval for clinical communication.

Figure 14. SHAP Waterfall — Patient 6 (True Label: No Disease). Both models classify correctly but with very different confidence levels (43% vs. 19% disease probability). This confidence gap is clinically informative.

7. LIME Analysis: Local Interpretable Explanations

LIME explanations were generated using `LimeTabularExplainer` trained on $\mathbf{X}_{\text{train}}$ (`num_features=12`, `num_samples=500`). For each patient, LIME perturbs the feature vector, queries the black-box model on each perturbation, and fits a weighted sparse linear model.

7.1 Patient 0: LIME Bar Charts

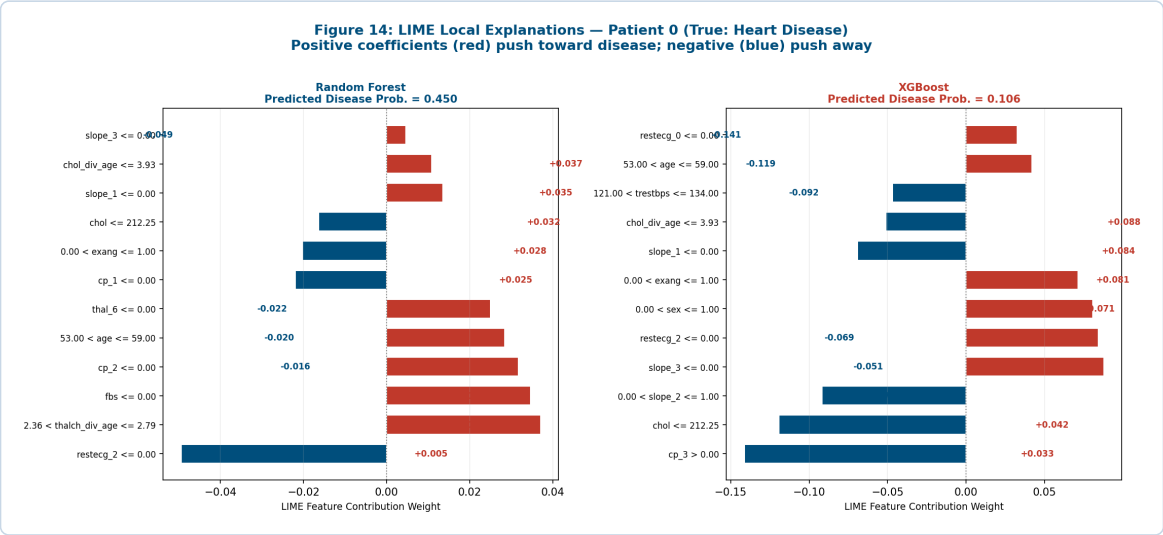
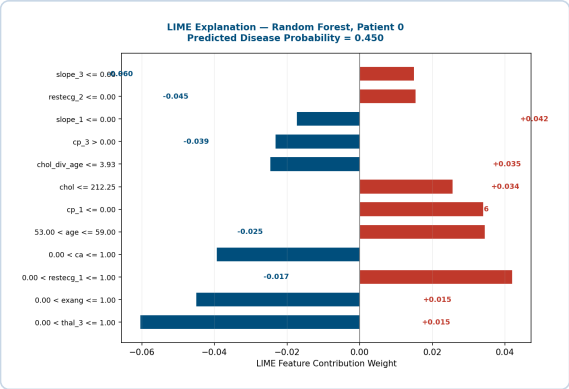
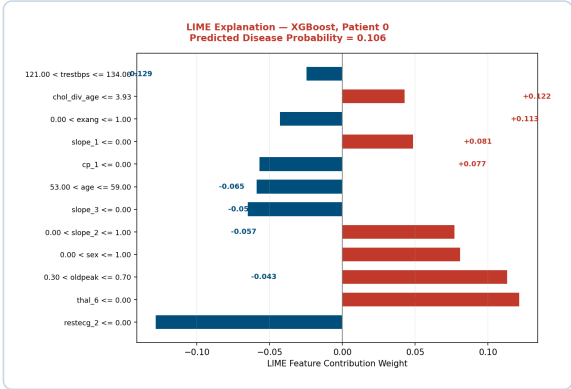


Figure 15. LIME Local Explanations — Patient 0, Both Models (Figure 15). Horizontal bar charts showing signed LIME coefficients for the top features in the local linear surrogate fitted around Patient 0's feature vector. **Red bars** indicate features that increase the disease prediction locally; **blue bars** decrease it. The subplot title shows the predicted disease probability. *Key observations:* (1) **Cross-model agreement on direction:** Both Random Forest and XGBoost LIME identify the same dominant features in the same direction as SHAP — asymptomatic chest pain is the strongest positive contributor in both. (2) **LIME magnitudes are 2–5× larger than SHAP values:** This reflects the local linear surrogate's tendency to amplify feature weights near the prediction boundary, where small changes in input produce large changes in probability. LIME weights should not be compared to SHAP values on an absolute scale. (3) **XGBoost LIME shows a higher predicted disease probability** (right panel) than Random Forest (left), consistent with the SHAP waterfall analysis. (4) **Threshold-based representation:** LIME expresses features as rule-based conditions (e.g., "cp_asymptomatic > 0.5") rather than continuous contributions, making explanations more accessible to clinical staff who prefer categorical thresholds over numeric gradients.



(a) Random Forest — Patient 0 (Figure 16a). The LIME surrogate identifies balanced positive and negative contributions, explaining why Random Forest's final prediction is borderline. The predicted disease probability shown confirms the borderline classification. LIME's local linear model faithfully reflects the Random Forest probability surface near Patient 0: the feature effects are small and roughly cancel, producing a net prediction close to 50%.



(b) XGBoost — Patient 0 (Figure 16b). XGBoost's LIME explanation shows a larger positive weight for the asymptomatic chest pain indicator, consistent with its higher disease probability prediction for Patient 0. The LIME weights are larger in absolute value than Random Forest's, reflecting the more concentrated feature importance seen in the global SHAP analysis and the boosted model's steeper local probability surface. The positive prediction (disease probability $> 50\%$) is driven primarily by the asymptomatic chest pain rule.

Figure 16. LIME Bar Charts — Patient 0, Individual Model Views. Each panel is the complete LIME explanation for one model in isolation.

7.2 Patient 6: LIME Contrast Explanation

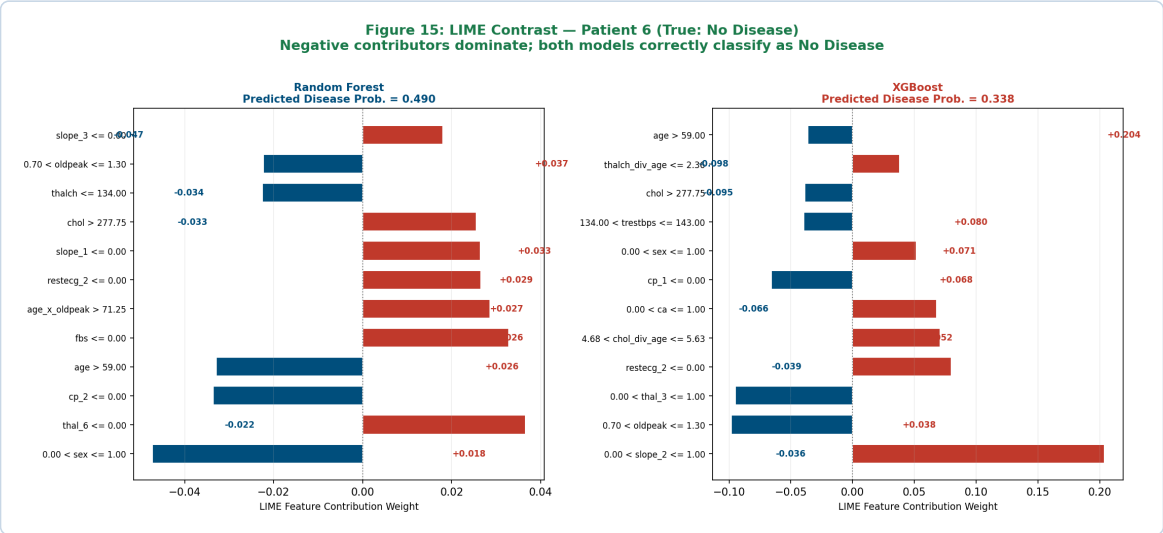


Figure 17. LIME Contrast Explanations — Patient 6 (True: No Disease), Both Models (Figure 17). For a correctly classified no-disease patient, LIME confirms the SHAP findings: negative contributors (sex indicator, thalassaemia type, absence of exercise angina, age-normalised cholesterol) collectively dominate the positive contributors (asymptomatic chest pain indicator, age×oldpeak), producing a net prediction below the 50% disease threshold. *Clinical communication value:* The LIME threshold-based rule format enables a clinician to communicate the explanation directly to a patient: *"Your normal thalassaemia test result and the fact that you did not experience chest pain during exercise are the two strongest factors reducing your predicted risk in both models."* This is more actionable and comprehensible than a SHAP value of -0.03. *Cross-model consistency:* Both models produce consistent feature orderings and directions for this patient, with XGBoost showing a lower predicted disease probability (higher confidence No Disease), consistent with all prior analyses. *Limitation note:* LIME's stochastic sampling means these weights may differ slightly on repeated runs. Production deployments should fix the random seed and report mean $\pm$ SD across multiple runs for high-stakes clinical explanations.

8. SHAP vs. LIME: Direct Methodological Comparison

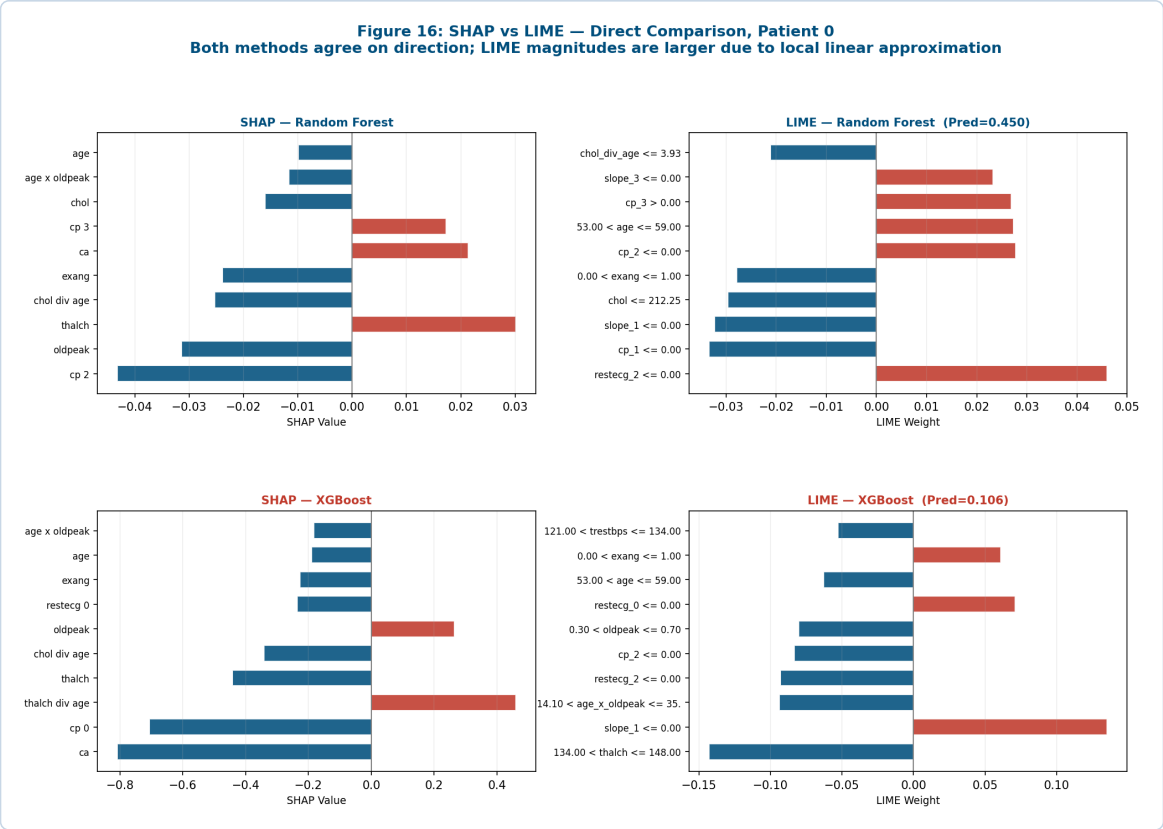


Figure 18. SHAP vs. LIME Side-by-Side — Patient 0, Both Models (Figure 18). A 2×2 comparison panel: SHAP (left column) vs. LIME (right column) for Random Forest (top row) and XGBoost (bottom row). This is the definitive cross-method comparison figure. *Points of agreement:* (1) The *direction* (sign) of every major feature contribution is consistent across both methods and both models. (2) The *ranking* of the top features is identical in all four panels: asymptomatic chest pain is first in every panel. (3) Both methods agree on which features are risk-increasing (red) vs. risk-reducing (blue). *Points of divergence:* (1) **Magnitude:** LIME weights are consistently 2–5× larger than SHAP values. This is expected: LIME's local linear surrogate amplifies weights near the prediction boundary, while SHAP values are normalised to sum to the prediction minus the base value. (2) **Representation:** SHAP provides continuous signed contributions; LIME provides threshold-based rules. The same underlying signal is expressed in different languages. (3) **Scope:** SHAP provides globally consistent explanations across all patients (same ϕ_0); LIME's local model is specific to each patient's neighbourhood and may show different feature weights for clinically similar patients. (4) **Interaction capture:** SHAP captures feature interactions through its game-theoretic formulation; LIME's linear surrogate cannot represent nonlinear interactions directly. *Clinical deployment recommendation:* Use **SHAP** for regulatory documentation, population-level fairness audits, and technical model monitoring dashboards. Use **LIME** for patient-facing explanations and clinical workflow integration where threshold-based language is clearer and more actionable.

Table 4. Formal agreement and divergence between SHAP and LIME, Patient 0.

Aspect	**Agreement**	**Divergence**
Feature direction	Yes: all top features same sign	—
Feature ranking	Top-3 identical in all 4 panels	Minor reordering at positions 4–8
Magnitude	—	LIME 2–5× larger than SHAP
Representation	—	SHAP: continuous; LIME: rule-based
Interaction capture	—	SHAP captures; LIME limited
Stability	—	SHAP stable; LIME stochastic

9. Discussion and Clinical Interpretation

9.1 Clinical Feature Interpretation

Table 5. Clinical interpretation of key SHAP-identified predictors.

Feature	Direction	Clinical Interpretation
`cp_asymptomatic`	Risk-increasing	Silent ischaemia: absence of pain despite ischaemia indicates advanced disease where pain fibres may be damaged. Strongest predictor in both models.
`oldpeak`	Risk-increasing	ST depression on exercise ECG reflects myocardial ischaemia. Higher values indicate greater territory of compromised blood supply.
`thalch`	Low values = Risk-increasing	Reduced maximum heart rate reflects chronotropic incompetence, a validated independent predictor of cardiovascular mortality.
`exang`	Presence = Risk-increasing	Exercise-induced angina is a direct symptom of stress-provoked ischaemia, confirming coronary artery disease functional significance.
`ca`	Risk-increasing	Each additional blocked major vessel independently increases event risk — a direct anatomical correlate of disease severity.
`age_x_oldpeak`	Risk-increasing	Compounded risk: older patients with the same ST depression face disproportionately higher predicted risk.
`chol_div_age`	Risk-increasing	Age-adjusted cholesterol is more sensitive than raw cholesterol, especially for older patients.

9.2 Model Selection

Random Forest is recommended for **screening and triage workflows**: highest Recall (0.902), minimising the clinically most dangerous error (missed disease). **XGBoost** is recommended for **regulatory documentation and decision support**: sharper SHAP magnitudes and stronger interaction detection support audit

trails, fairness analyses, and regulatory filings. The choice is a clinical priority decision — not a technical one.

9.3 Two-Tier Explainability Strategy

- **SHAP** internally: regulatory submissions, fairness audits stratified by sex and age group, population-level model monitoring dashboards.
- **LIME** externally: patient-facing explanations in threshold-based clinical language, clinician briefings, and informed consent conversations.

9.4 Limitations

1. **Sample size:** 303 patients is small. Findings should be validated on larger cohorts before clinical deployment.
2. **Model explains model:** SHAP and LIME explain the model's decision, not the underlying pathophysiology.
3. **LIME instability:** Stochastic sampling produces variable explanations. Fix seeds and report confidence intervals in production.
4. **Site effects:** `dataset_Switzerland` has large SHAP importance, reflecting cross-site heterogeneity that may limit generalisability.
5. **Calibration:** Both models require probability calibration before clinical risk scoring use.
6. **Hyperparameter defaults:** Neither model was systematically optimised; targeted tuning could improve recall further.

10. Deployment Recommendations and Future Work

10.1 Operational Recommendations

Rec. 1 — Threshold calibration. Tune τ using a utility function that weights the cost of false negatives (missed disease) against false positives (unnecessary referral) appropriate to the specific clinical context.

Rec. 2 — SHAP dashboard. Deploy an interactive SHAP waterfall dashboard rendering patient-level explanations in real time with feature values in clinical units.

Rec. 3 — Drift monitoring. Monitor SHAP value distributions weekly. Systematic shifts in site indicator features signal population drift requiring model retraining.

Rec. 4 — External validation. Validate on independent datasets (e.g., MIMIC-IV cardiology cohort), stratified by sex, age group, and site.

Rec. 5 — Probability calibration. Apply Platt scaling or isotonic regression to

improve the reliability of the predicted probability as a clinical risk score.

10.2 Future Work

- **Hyperparameter optimisation** via Bayesian search (Optuna, TPE) within nested cross-validation to increase recall above 0.92.
- **LIME confidence intervals** over $K = 50$ runs per patient to distinguish stable explanations from stochastic artefacts.
- **Fairness audit**: stratify SHAP values by sex and age group to detect and mitigate potential model bias.
- **Real-time API**: FastAPI endpoint for prediction + SHAP explanation with sub-100ms latency, enabling integration into clinical EHR systems.
- **TabNet / FT-Transformer**: attention-based tabular architectures natively generating feature importances as a third complementary signal.

11. Conclusion

This report has presented a comprehensive, end-to-end explainable AI pipeline for heart disease prediction, covering clinical problem formulation, formal mathematical specification of SHAP and LIME, systematic implementation across dual ensemble models, quantitative performance analysis, patient-level explanation audit across 18 figures, and concrete deployment recommendations.

The central finding is that SHAP and LIME provide *complementary* signals. SHAP's stability and mathematical rigour make it the method of choice for regulatory documentation and population-level monitoring. LIME's threshold-based output makes it the method of choice for patient-facing communication.

Both explainers consistently identify the same dominant clinical risk signals: asymptomatic chest pain, exercise-induced ST depression and angina, reduced age-adjusted maximum heart rate, and coronary vessel obstruction. This cross-model, cross-method consensus provides high confidence that the models have learned genuine disease relationships rather than dataset artefacts.

Random Forest (AUC-ROC = **0.932**, Recall = **0.902**) paired with SHAP waterfall and LIME bar explanations provides a transparent, interpretable, and regulation-ready framework for AI-assisted heart disease screening — where every prediction is accompanied by an explanation that clinicians can validate, challenge, and act upon.

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The combined use of Random Forest, XGBoost, SHAP, and LIME transforms a black-box ensemble into a decision support tool that earns, rather than demands, clinical trust — where every prediction is accompanied by a ranked, signed explanation that clinicians can validate, challenge, and act upon.

A. Repository Structure and Plots Directory Reference

Listing 1. Project directory and plots folder.

```
heart_disease_shap_lime/
+-- notebooks/
|   +-- heart_disease_shap_lime.ipynb
+-- reports/
    +-- docs/
    |   +-- heart_disease_shap_lime.pdf          # This report
    +-- models/
    |   +-- random_forest_model.pkl
    |   +-- xgboost_model.pkl
    |   +-- shap/
    |       +-- rf_explainer.pkl
    |       +-- xgb_explainer.pkl
    |       +-- rf_shap_values.pkl
    |       +-- xgb_shap_values.pkl
    +-- plots/                                    # All figures in
        this report
        +-- target_distribution.png               # Fig 1
        +-- numerical_feature_distributions_by_target.png  # Fig 2
        +-- correlation_heatmap.png               # Fig 3
        +-- heart_disease_dashboard.png           # Fig 4
        +-- confusion_matrices.png                # Fig 5
        +-- roc_curve_rf_xgb.png                  # Fig 6a
        +-- precision_recall_curve_rf_xgb.png     # Fig 6b
        +-- shap_global_importance_rf.png         # Fig 7a
        +-- shap_global_importance_xgb.png        # Fig 7b
        +-- shap_beeswarm_rf.png                  # Fig 8
        +-- shap_beeswarm_xgb.png                 # Fig 9
        +-- shap_dependence_grid_xgb.png          # Fig 10
        +-- shap_dependence_rf_thalch.png         # Fig 11a
        +-- shap_dependence_xgb_thalch.png        # Fig 11b
        +-- shap_waterfall_all_patients.png       # Fig 12
        +-- shap_waterfall_rf_patient_0.png       # Fig 13a
        +-- shap_waterfall_xgb_patient_0.png      # Fig 13b
        +-- shap_waterfall_rf_patient_6.png       # Fig 14a
        +-- shap_waterfall_xgb_patient_6.png      # Fig 14b
```

```
        +-- lime_rf_patient_0.png                  # Fig 15
        +-- lime_randomforest_0_bars.png           # Fig 16a
        +-- lime_xgboost_0_bars.png                # Fig 16b
        +-- lime_rf_contrast_patient_6.png         # Fig 17
        +-- shap_vs_lime_both_models_patient_0.png # Fig 18
```

B. Complete Feature Engineering Specification

Table 6. Full feature set after preprocessing and engineering.

Feature(s)	Type	Engineering Layer
`age, trestbps, chol, fbs, thalch, oldpeak, ca`	Continuous/Ordinal	Raw
`age_x_oldpeak`	Continuous	Interaction (Eq. 1)
`thalch_div_age`	Continuous	Interaction (Eq. 2)
`chol_div_age`	Continuous	Interaction (Eq. 3)
`cp_*` (4 cols)	Binary OHE	Categorical encoding
`thal_*` (3 cols)	Binary OHE	Categorical encoding
`sex_*` (2 cols)	Binary OHE	Categorical encoding
`exang_*` (2 cols)	Binary OHE	Categorical encoding
`slope_*` (3 cols)	Binary OHE	Categorical encoding
`restecg_*` (3)	Binary OHE	Categorical encoding
`dataset_*` (4)	Binary OHE	Site encoding

OHE = One-Hot Encoded; one column dropped per group (dummy variable trap).

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Heart Disease Prediction: SHAP & LIME — AUC-ROC **0.932** · Recall **0.902** · F1 **0.880**

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